

Generalized method of a resolvent operator expansion. II.

M. Znojil

Institute of Nuclear Physics, Czechoslovak Academy of Sciences, 250 68 Řež, Czechoslovakia
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The second part of the present series is devoted to the rearrangement of the Born series in the model space. An application is found in pion-nucleus scattering. We present a generalization of the "extended schematical model" suggested recently by Hirata *et al.*

1. INTRODUCTION

Inversion of a linear operator which is not necessarily Hermitian is a typical mathematical operation used in the scattering theory. The well-known Born expansion of a resolvent operator being not applicable in the whole region of energies, other approximation methods should be used in the intermediate energy region. In the present paper, we are inspired by the recent development in pion-nucleus scattering physics and describe the method which, within a model space, performs an efficient analytic continuation of high energy expansion by representing it in an equivalent matrix continued fraction form.

The method is obtained from the abstract parametric expansion of a resolvent $R(E) = (E - H)^{-1}$ as described in Ref. 1 (denoted I). As a new feature, the knowledge of the matrix elements of powers of H (moments) is required between the bra and ket vectors of the model spaces $d \subset V^*$ and $D \subset V$, respectively. This enables one to calculate explicitly the input parameters for definition of the expansion I.

In Sec. 2, we define the basis in the whole Hilbert space by applying the general parametrization¹ to H and H^* separately and requiring block biorthogonality. In the special one-dimensional case, the resulting biorthogonal basis coincides with the one often used in numerical practice (Lanczos²) and physical calculations (Haydock and Kelly³).

Although the whole resolvent expansion I may be written immediately, the knowledge of scalar product (not employed in I) makes possible an efficient approximation technique. Due to the variability of the model spaces and their dimension M , all the necessary matrix elements of $R(E)$ may be taken between d and D without loss of generality. All the remaining basis vectors $\notin d, D$ are eliminated in Sec. 3, and their use is replaced by the knowledge of moments. This algebraic rearrangement leads to an efficient algorithm representing $R(E)$ between d and D by the matrix continued fraction formula. In the special case $M = 1$, we obtain the classical continued fraction results.⁴

The physical and numerically tested example of applicability of the present method is the extended schematical model (ESM) of the elastic π -⁴He and π -¹⁶O scattering given by Hirata *et al.*^{5,6} It is described in Sec. 4 and shown to be equivalent to the one-dimensional version of the present method. This is the heuristic basis for the coupled channel generalization of ESM: We suggest that due to the analogy of

the physical (isobar doorway) picture, the similar cutoff ($N \sim 3-4$) of the continued fraction (with $M > 1$) should work in the description of inelastic pion-nucleus scattering, with new indices representing the different coupled channels. The detailed numerical calculation should be given elsewhere, as it may vary in dependence on additional physical assumptions.

2. GENERALIZED LANCZOS BASIS

We shall work in the vector space V and its dual V^* and employ small or capital letters for denoting bras ($\langle y | \in V^*$) or kets ($| Y \rangle \in V$), respectively. The kets (bras) will always occur in groups numbered by a pair of indices, e.g., $| X_k^i \rangle$, $i = 1, 2, \dots, M_k$ ($i \in \overline{1, M_k}$), $k = 1, 2, \dots$. The standard matrix conventions with respect to these indices will be accepted (kets or bras being ordered in rows or columns, respectively) so that we shall be allowed to omit either upper indices or both of them [compare respectively Eqs. (2) and (3) or Eqs. (3) and (4) below].

Let us consider a linear operator H on V . We shall formulate our task as the evaluation of the matrix elements of the type

$$T^{ij} = \langle x_i | R(E) | X_j^i \rangle, \quad (1)$$

$$R(E) = (E - H)^{-1}, \quad i, j \in \overline{1, M_1}$$

for some arbitrarily chosen finite set of bras $\langle x_i |$ and an independent set of kets $| X_j^i \rangle$ spanning the model spaces $d \subset V^*$ and $D \subset V$ respectively, $d \neq D$ in general. In the present section we describe the introduction of the block-biorthogonal basis generated from $\langle x_i |$ and $| X_j \rangle$.

We employ the formalism for expanding the resolvent operator $R(E)$ as described in Ref. 1. First, the two sets of groups of vectors

$$| X \rangle = \{ \{ | X_k^i \rangle, i \in \overline{1, M_k} \}, k = 1, 2, \dots \}$$

and

$$\langle x \rangle = \{ \{ \langle x_k^i |, i \in \overline{1, m_k} \}, k = 1, 2, \dots \}$$

will be defined in a recurrent way from the initial groups $| X_1 \rangle$ and $\langle x_1 |$. According to I, the group $| X_{k+1} \rangle$ will be determined for any $k \geq 1$ by the k -fold action of H on $| X_1 \rangle$. The partial subtraction of the linear combinations of "old group" vectors is also allowed by the corresponding recurrent defi-

inition (we slightly change the notation of I)

$$H|X_k^i\rangle = \sum_{j=1}^{M_k} |X_{k-1}^j\rangle C_k^{ij} + \sum_{j=1}^{M_k} |X_k^j\rangle A_k^{ij} + \sum_{j=1}^{M_{k+1}} |X_{k+1}^j\rangle B_k^{ij} \quad (2)$$

with arbitrary groups of parameters A, B, C . The shortened notation enables us to abbreviate Eq. (2) as

$$H|X_k\rangle = |X_{k-1}\rangle C_k + |X_k\rangle A_k + |X_{k+1}\rangle B_k \quad (3)$$

or even

$$H|X\rangle = |X\rangle Q. \quad (4)$$

The same sort of parametrization will be used for the action of the Hermitian conjugate operator H^* ($\neq H$ in general) on another initial group $|x_1\rangle \in V$, $H^*|x\rangle = |x\rangle q^*$, i.e., $\langle x|H = q\langle x|$, i.e.,

$$\langle x_k^i|H = \sum c_k^{ij}\langle x_{k-1}^j| + \sum a_k^{ij}\langle x_k^j| + \sum b_k^{ij}\langle x_{k+1}^j|. \quad (5)$$

Employing formulas of I, we shall be able to write formally the expansion of $R(E)$ or $R^*(E)$. Once the scalar products $\langle x_k^i|X_l^j\rangle = D_{kl}^{ij}$ are known, both parametrizations of Eqs. (4) and (5) are not independent and may be used together as an input for I due to the identities

$$\langle x|H|X\rangle = DQ = qD. \quad (6)$$

In our paper, we reduce the complexity of the general expansion by demanding the block-diagonal form of D :

$$\begin{aligned} \langle x_k^i|X_l^j\rangle &= D_{kl}^{ij}, \quad k=l, \quad m_k = M_k, \\ &= 0, \quad k \neq l. \end{aligned} \quad (7)$$

This assumption (block-biorthogonality) will naturally restrict the freedom in the parametrizations (4) and (5) since Eq. (6) implies

$$\begin{aligned} \langle x_{k-1}|H|X_k\rangle &= D_{k-1,k}C_k = b_{k-1}D_k, \\ \langle x_k|H|X_k\rangle &= D_{k,k}A_k = a_kD_k, \\ \langle x_{k+1}|H|X_k\rangle &= D_{k+1,k}B_k = c_{k+1}D_k. \end{aligned} \quad (8)$$

This is very analogous to the one-dimensional case⁷ ($M=1$) where the sets $|X\rangle$ and $\langle x|$ will form the well-known Lanczos biorthogonal basis.²

To restrict the freedom in the parametrizations (4), (5) allowed by Eq. (8) and hidden in the arbitrary decomposition of the linear superposition $b_k\langle x_{k+1}|$ and $|X_{k+1}\rangle B_k$, the modified vectors may be introduced in the form

$$\begin{aligned} |\Xi_{k+1}\rangle &= |X_{k+1}\rangle \Delta_k, \\ \Delta_k &= B_k B_{k-1} \dots B_1, \quad \Delta_0 = I, \\ \langle \xi_{k+1}| &= \delta_k \langle x_{k+1}|, \\ \delta_k &= b_1 b_2 \dots b_k, \quad \delta_0 = I. \end{aligned} \quad (9)$$

Assuming

$$\begin{aligned} \det B_k \neq 0 \neq \det b_k, \quad 0 \neq \det D_k, \\ M_k = M, \quad k = 1, 2, \dots \end{aligned} \quad (10)$$

the vectors of Eq. (9) are defined uniquely by the recurrences

$$\begin{aligned} H|\Xi_k\rangle &= |\Xi_{k-1}\rangle g_{k-1} \beta_k + |\Xi_k\rangle g_k \alpha_k + |\Xi_{k+1}\rangle, \\ \langle \xi_k|H &= \beta_k g_{k-1} \langle \xi_{k-1}| + \alpha_k g_k \langle \xi_k| + \langle \xi_{k+1}|, \end{aligned} \quad (11)$$

$$\beta_k = \delta_{k-1} D_k \Delta_{k-1} = (g_k)^{-1}, \quad g_0 = 0,$$

$$\alpha_k = \delta_{k-1} a_k D_k \Delta_{k-1},$$

which follow from Eqs. (4), (5), (10). From the formal point of view Eqs. (11) may be considered to be special cases of parametrization (4), (5) with $B = I = b$. The resolvent operator expansion formulas I may thus be summarized by the expression

$$\begin{aligned} \langle \xi_k|R(E)|\Xi_m\rangle &= \sum_{l=1}^{\min(k,m)} [(\varphi_k g_{k-1})(\varphi_{k-1} g_{k-2}) \dots (\varphi_{l+1} g_l)] \\ &\quad \times \varphi_l [(g_l \varphi_{l+1}) \dots (g_{m-1} \varphi_m)], \end{aligned} \quad (12)$$

where $[\dots] = I$ for $k=l$ or $l=m$ and the backward matrix recurrence

$$\varphi_k = (E g_k - g_k \alpha_k g_k - g_k \varphi_{k+1} g_k)^{-1}, \quad \varphi_{N+1} = 0, \quad (13)$$

defines the auxiliary sequence φ_k in terms of α, β and $N \gg 1$ ($N \rightarrow \infty$).

3. MODEL SPACE DEFINITION OF PARAMETERS

In this section, we shall generalize some relations known for $M=1$ in the classical theory of moments.⁴ We shall eliminate the higher groups of vectors ($k>1$) from the definitions of type (8) and arrive at analogs of the classical relations expressing, e.g., B_k of the one-dimensional case as a square root of the ratio of determinants formed from the Hankel matrices of momenta.⁴

By inserting the vectors (9) given by Eq. (11) into the matrices

$$\begin{aligned} \langle \xi_{k+1}|H^l|\Xi_{k+1}\rangle &= \alpha_{k+1}^{[l]}, \quad \langle \xi_{k+1}|H^l|\Xi_k\rangle = \gamma_k^{[l]}, \\ \langle \xi_k|H^l|\Xi_{k+1}\rangle &= \beta_{k+1}^{[l]}, \end{aligned} \quad (14)$$

we obtain the recurrence relations for matrices (14) initialized at $k=0$ by the (generalized) moments ($M_1 \times M_1$ matrices) $\alpha^{[l]}$ denoted as

$$\langle \xi_l|H^l|\Xi_l\rangle = \langle x_l|H^l|X_l\rangle = \langle l\rangle, \quad l = 0, 1, 2, \dots \quad (15)$$

Hence the two sets of matrices $\alpha_k = \alpha_k^{[1]}, \beta_k = \beta_k^{[1]} = \gamma_{k-1}^{[1]}$ are uniquely defined by H and by the choice of groups $|X_l\rangle$ and $\langle x_l|$ spanning model spaces D and d , respectively. It is not surprising, since the simple expansion of $R(E)$ within the model space $[\equiv \varphi_1(E)]$ into powers of H provides the same conclusion.

The insertion of α_k, β_k into Eq. (13) defines φ_1 , i.e., the rearrangement of the standard power series in form of the matrix continued fraction. Although the convergence is not investigated now, we shall assume that it is sufficiently quick in the practical calculations (cf., e.g., Sec. 4 or Ref. 8).

The recurrences resulting from Eq. (14) do not give the optimal form of the desired matrices since many cancellations occur. In spite of the cancellations, they may be used for $k = 1, 2$, and 3 , given

$$\begin{aligned}\beta_1 &= g_1^{-1} = D_1 = \langle 0 \rangle, \quad \alpha_1 = \langle 1 \rangle, \\ \beta_2 &= g_2^{-1} = \langle 2 \rangle - \langle 1 \rangle g_1 \langle 1 \rangle, \\ \alpha_2 &= \langle 3 \rangle - \langle 2 \rangle g_1 \langle 1 \rangle - \langle 1 \rangle g_1 \langle 2 \rangle + \langle 1 \rangle g_1 \langle 1 \rangle g_1 \langle 1 \rangle, \\ \beta_3 &= \langle 4 \rangle - \langle 2 \rangle g_1 \langle 2 \rangle - (\langle 3 \rangle - \langle 2 \rangle g_1 \langle 1 \rangle) \\ &\quad \times g_2 (\langle 3 \rangle - \langle 1 \rangle g_1 \langle 2 \rangle), \\ \alpha_3 &= \langle 5 \rangle - \langle 2 \rangle g_1 \langle 1 \rangle g_1 \langle 2 \rangle - [(\langle 3 \rangle - \langle 2 \rangle g_1 \langle 1 \rangle) g_2 (\langle 4 \rangle \\ &\quad - \langle 1 \rangle g_1 \langle 1 \rangle g_1 \langle 2 \rangle) - [\text{h.c.}] + (\langle 3 \rangle - \langle 2 \rangle g_1 \langle 1 \rangle) \\ &\quad \times g_2 (\langle 3 \rangle - \langle 1 \rangle g_1 \langle 1 \rangle g_1 \langle 1 \rangle) g_2 (\langle 3 \rangle - \langle 1 \rangle g_1 \langle 2 \rangle)].\end{aligned}\quad (16)$$

Because of the rapidly increasing complexity of these expressions, we prefer to simplify them by introducing some auxiliary subexpressions for higher k .

We start with the definition of projectors and operators

$$\begin{aligned}P_k &= |\Xi_k\rangle g_k \langle \xi_k|, \quad S_k = S_{k-1} - P_k, \\ k &= 1, 2, \dots, \quad S_0 = I, \\ \Omega_k &= 1 - |\Xi_{k+1}\rangle g_{k+1} \langle \xi_{k+1}| H^k, \\ \omega_k &= 1 - H^k |\Xi_1\rangle g_{k+1} \langle \xi_{k+1}|.\end{aligned}\quad (17)$$

Exploiting the block tridiagonality of Q in Eq. (4) and employing the obvious properties

$$\begin{aligned}|\Xi_k\rangle &= S_{k-1} H |\Xi_{k-1}\rangle = S_{k-1} H S_{k-2} H |\Xi_{k-2}\rangle \\ &= S_{k-1} H^{k-1} |\Xi_1\rangle,\end{aligned}\quad (18)$$

$$\langle \xi_k | = \langle \xi_1 | H^{k-1} S_{k-1}, \quad \Omega_k |\Xi_{k+1}\rangle = 0 = \langle \xi_{k+1} | \omega_k,$$

$$S_{k+1} = \Omega_k S_k = S_k \omega_k,$$

we arrive at the fundamental implicit form of the recurrent definitions of parameters

$$\begin{aligned}\beta_k &= \langle \xi_k | \Xi_k \rangle = \langle \xi_1 | H^{k-1} S_{k-1} H^{k-1} |\Xi_1\rangle, \\ \alpha_k &= \langle \xi_1 | H^{k-1} S_{k-1} H S_{k-1} H^{k-1} |\Xi_1\rangle,\end{aligned}\quad (19)$$

$$S_{k-1} = S_{k-2} - S_{k-2} H^{k-2} |\Xi_1\rangle g_{k-1} \langle \xi_{k-1}| H^{k-2} S_{k-2}.$$

To derive the explicit algorithm which should give an iterative definition of β_k , based on the initialization (15), we introduce the auxiliary sequence $\eta_{k+1}^{[l,r]}$, $l, r \geq k \geq 1$, defined by the recurrence

$$\eta_{k+1}^{[l,r]} = \eta_k^{[l,r]} - \eta_k^{[l,k-1]} g_k \eta_k^{[k-1,r]}, \quad \eta_1^{[l,r]} = \langle l+r \rangle, \quad (20)$$

which coincides with the required sequence β_{k+1} for $l=r=k$ since $\eta_{k+1}^{[l,r]} = \langle \xi_1 | H^l S_k H^r |\Xi_1\rangle$, $k \geq 0$, $\eta_{k+1}^{[k,k]} = \beta_{k+1}$. Thus, Eq. (20) is the first part of the desired algorithm.

The form of α_k will be more complex since the definition (19) contains two projectors. By the explicit use of the projection properties of Ω_k and ω_k , it is a rather trivial block-matrix multiplication manipulation to show that the ansatz

$$\alpha_{k-1} = \langle \xi_1 | H^{k-2} \Omega_{k-3} Z_{k-3} \times \omega_{k-3} H^{k-2} |\Xi_1\rangle, \quad k = 3, 4, \dots, \quad (21)$$

$$Z_0 = H, \quad Z_1 = H - P_1 H P_1, \dots,$$

implies

$$\begin{aligned}\alpha_{k+1} &= \langle \xi_1 | H^k \Omega_{k-1} [\Omega_{k-3} Z_{k-3} \omega_{k-3} \\ &\quad - P_{k-1} H P_{k-1}] \omega_{k-1} H^k |\Xi_1\rangle, \quad k = 3, 4, \dots.\end{aligned}$$

It enables us to write

$$\begin{aligned}\alpha_{k+1} &= (\langle \xi_1 | H^k - \langle \xi_1 | H^k S_{k-1} H^{k-1} |\Xi_1\rangle g_k \langle \xi_1 | H^{k-1} \rangle Z_{k-1} \\ &\quad \times (H^k |\Xi_1\rangle - H^{k-1} |\Xi_1\rangle) g_k \langle \xi_1 | H^{k-1} S_{k-1} H^k |\Xi_1\rangle)\end{aligned}\quad (22)$$

[cf. Eq. (21), $k > 1$] and derive the recurrence for operators Z

$$\begin{aligned}Z_k &= \Omega_{k-2} (Z_{k-2} - S_{k-2} H^{k-1} |\Xi_1\rangle g_k \\ &\quad \times \alpha_k g_k \langle \xi_1 | H^{k-1} S_{k-2} \rangle \omega_{k-2}), \quad k = 2, 3, \dots\end{aligned}\quad (23)$$

From this we define the matrices $\vartheta_k^{[l,r]} = \langle \xi_1 | H^l Z_k H^r |\Xi_1\rangle$, $l, r \geq k$ which, due to Eq. (23), satisfy the recurrent definition

$$\begin{aligned}\vartheta_k^{[l,r]} &= \mu_{k-2}^{[l,r]} - \eta_{k-1}^{[l,k-2]} g_{k-1} \mu_{k-2}^{[k-2,r]} \\ &\quad - \mu_{k-2}^{[l,k-2]} g_{k-1} \eta_{k-1}^{[k-2,r]} + \eta_{k-1}^{[l,k-1]} g_{k-1} \\ &\quad \times \mu_{k-2}^{[k-2,k-2]} g_{k-1} \eta_{k-1}^{[k-2,r]}, \quad k = 3, 4, \dots,\end{aligned}\quad (24)$$

where $\mu_{k-2}^{[l,r]} = \vartheta_{k-2}^{[l,r]} - \eta_{k-1}^{[l,k-1]} g_k \alpha_k g_k \eta_{k-1}^{[k-1,r]}$ and the initializations are

$$\begin{aligned}\vartheta_0^{[l,r]} &= \langle l+r+1 \rangle, \\ \vartheta_1^{[l,r]} &= \langle l+r+1 \rangle - \langle l \rangle g_1 \alpha_1 g_1 \langle r \rangle, \\ \vartheta_2^{[l,r]} &= \langle \xi_1 | H^l S_1 H S_1 H^r |\Xi_1\rangle - \eta_2^{[l,1]} g_2 \alpha_2 g_2 \eta_2^{[1,r]}.\end{aligned}\quad (25)$$

From Eq. (22) we obtain the definition

$$\begin{aligned}\alpha_{k+1} &= \vartheta_{k-1}^{[k,k]} - \eta_k^{[k,k-1]} g_k \vartheta_{k-1}^{[k-1,k]} \\ &\quad - \vartheta_{k-1}^{[k,k-1]} g_k \eta_k^{[k-1,k]} + \eta_k^{[k,k-1]} g_k \vartheta_{k-1}^{[k-1,k-1]} \\ &\quad \times g_k \eta_k^{[k-1,k]}, \quad k = 1, 2, \dots, \quad \alpha_1 = \langle 1 \rangle,\end{aligned}\quad (26)$$

of the parameter matrices α_{k+1} , i.e., the second part of the algorithm based on the second auxiliary sequence $\vartheta_k^{[l,r]}$ defined by the recurrence (24). Thus, Eq. (13) with $k = N$, $N = 1, \dots, 1$ and Eqs. (20), (24), and (26) define the matrix elements of $R(E)$ between the vectors from d and D model subspaces in terms of moments $\langle l \rangle$, $l = 0, 1, \dots, 2N-1$.

4. AN EXAMPLE OF APPLICATION: ESM EXTENDED

The role of relativistic effects in the pion-nucleus scattering is probably not negligible, and some of the recent attempts try therefore to introduce the isobar degrees of freedom into the nucleus and employ the dominant role of the π - N (1232) resonance in the microscopic description of the π - ^4He ^{5,9} and π - ^{16}O ⁶ elastic scattering at intermediate energies. We shall briefly review the algorithm suggested by Hirata *et al.*⁶ (ESM) since it considers the pion-nucleus transition operator just in the form of Eq. (1) with $M = 1$. The $|X_1\rangle$ or $\langle x_1|$ state is assumed to be a coherent superposition of the isobar-hole (Δh) states representing the response of the nuclear ground state to the incoming pion of momentum k , $|X_1\rangle = \sum |\Delta h\rangle \langle \Delta h | H | k, K \rangle$, $K = 0$, or outgoing pion of momentum k' , $\langle x_1| = \sum \langle k', K | H | \Delta h \rangle \langle \Delta h |$, $K = 0$, respectively. We might therefore apply our method with $M = 1$. In Ref. 6 the equivalent numerical evaluation is fulfilled along the standard lines: The complete basis $|e_1\rangle, |e_2\rangle, |e_3\rangle, \dots, |e_{\bar{N}}\rangle$ in V is constructed by using the Lanczos method⁷ and diagonalizing the realistic semiphenomenological Hamiltonian H . The inversion (Green function) is then given by summing all the contributions from the $\bar{N}$ separate states.

The numerical conclusions of Ref. 6 are surprising and nontrivial: The cutoff parameter $N \ll \bar{N}$ sufficient to accurate description of π - ^{16}O elastic scattering turned out to be as low as 3. This defines ESM and restricts strongly the necessary number N of collective doorway states $|e_k\rangle$ (linear combinations of $|X_1\rangle, |X_2\rangle, \dots, |X_N\rangle$) belonging to the whole isobar-hole space V .

The same result has been obtained for different nuclei.⁵ This inspires us to the assumption that also the nonelastic scattering may be considered along the same lines. Using the same physical picture of the isobar excitations in the resonance region, the adequate method should be based on our matrix continued fraction formula (13) with $k = 1$ which is entirely equivalent to diagonalization due to the use of the same truncated part of the vector space V spanned by the groups $|X_1\rangle, \dots, |X_N\rangle$.

Even in the simplest one-dimensional version, our method represents further simplification of ESM since we have eliminated the redundant intermediate step of diagonalization which is an infinite numerical process in principle.⁷ Using $M > 1$, we may take into account the possibility of channel coupling reflected by interaction within the k th

group $|X_k^i\rangle$, $k \geq 1$, with the upper index numbering the channels. Assuming that the physical mechanism (isobar creation) is not modified too much in the inelastic processes, we are entitled to use even the same level of approximation ($N = 3-4$), although more involved physical structure is described by more states, channel coupling, and $M > 1$. The algebraic utilization of the block tridiagonality of H in block-biorthogonal basis is related also to certain numerical advantages of the continued fraction form.⁴ The detailed evaluations of the pi-nucleus cross sections are to be published elsewhere after clarifying the problems connected with the proper choice of the effective interaction H .^{5,9} This is out of the scope of the present paper.

5. SUMMARY

While in I the series expansion of the vectors $R(E)|X_1^i\rangle$ in terms of the expansion set $|X_k\rangle \sim H^{k-1}|X_1\rangle + \dots$, $k = 1, 2, \dots$, was derived, the present paper is oriented to the expansion of the model space matrix elements $\langle x_1^i | R(E) | X_1^j \rangle$ into the matrix continued fraction by eliminating all the groups of vectors $\langle x_k |$, $|X_k\rangle$ with $k > 1$. The recurrences are given which define the parameters of the continued fraction in terms of moment matrices $\langle x_1^i | H^l | X_1^j \rangle$ in the most compact way. The groups of $M \geq 1$ vectors $\langle x_1 |$ and $|X_1\rangle$ span two independent bra and ket model spaces d and D , respectively. This guarantees the broad and flexible applicability of the method which was illustrated by simplifying and generalizing ESM⁶ in pion-nucleus scattering.

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